

Pratik Jadhav

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SUMMARY

AI/ML Engineer with hands-on experience building production ML systems, LLM applications, and distributed AI backends. Reduced inference latency by 78% in a production LLM pipeline and improved manufacturing defect detection accuracy from 34% to 92%. Experienced designing and deploying AI systems across computer vision and generative AI.

TECHNICAL SKILLS

Languages: Python, SQL

AI/ML: LLMs, RAG, Transformers, CNNs, PyTorch, TensorFlow, Scikit-learn, Hugging Face, DSPy, LangGraph

Backend: FastAPI, REST APIs, WebSockets, Celery, RabbitMQ, Redis, Async Python

Retrieval & Databases: FAISS, BM25, PostgreSQL, MongoDB, MySQL

Cloud & MLOps: Docker, MLflow, Vertex AI, TensorFlow Serving, GitHub Actions

PROFESSIONAL EXPERIENCE

AI/ML Engineer, Joshnik AI Labs

03/2025 – Present | Bangalore, India

- Engineered a data pipeline to crawl, classify, and index 50K+ business profiles across 48 categories, improving classification coverage through hierarchical fallback routing.
- Optimized a DSPy-powered fashion intelligence agent serving 16K+ brands by parallelizing retrieval, replacing LLM pre-filtering with regex ranking, and limiting retrieval context, reducing end-to-end latency from 18s to 4s (78%).
- Developed an asynchronous real estate data platform where FastAPI orchestrates Celery workers via RabbitMQ to crawl listings, extract structured property data using LLMs, persist results to MongoDB, and stream processing status to a live dashboard.
- Built an LLM-powered extraction pipeline using Crawl4AI and DSPy to convert raw HTML into structured property data, supporting multiple LLM providers through a unified inference interface.
- Architected multi-tenant marketplace infrastructure integrating payments, real-time slot reservations, and meeting scheduling for consumer and professional workflows.

Machine Learning Intern, Alemeno Private Limited

04/2024 – 10/2024 | Mumbai, India

- Developed an LCD defect detection system covering 17 defect classes using an ensemble of CNN and LLM models, improving production accuracy from 34% to 92% while reducing false positives.
- Constructed and manually annotated a 10K+ image dataset with bounding-box labels, enabling training of production-grade defect detection models deployed via Vertex AI and Dockerized TensorFlow Serving.
- Improved difficult-class detection accuracy from 54% to 81% through targeted hard-negative mining and iterative model retraining.

Applied AI Intern, Atria University

08/2023 – 03/2024 | Bangalore, India

- Designed an LLM-as-a-Judge evaluation framework for a RAG-based agricultural assistant, improving retrieval precision by 25–35% while reducing hallucinations through retrieval and prompt optimization.

Machine Learning Intern, Cheslab Private Limited

04/2023 – 06/2023 | Bangalore, India

- Developed a CNN-based skin disease classifier across 12 disease categories using 100K+ images, leading the end-to-end data pipeline and integrating the model into a production diagnosis workflow.

PROJECTS

Curalink — AI Medical Research Assistant | 3rd Place Winner, AI Hackathon (800+ Participants) 

FastAPI · Flan-T5 · LoRA · Groq Llama-3.3-70B · MiniLM · React · MongoDB

- Fine-tuned Flan-T5 using LoRA on 8K medical examples to translate natural-language queries into PubMed MeSH Boolean searches, publishing the model to Hugging Face.
- Built a hybrid RAG pipeline over PubMed, OpenAlex, and ClinicalTrials.gov, reranking candidates using MiniLM embeddings before synthesizing grounded responses with Llama-3.3-70B.

Facial Emotion-Based Music Recommendation System 

TensorFlow · Keras · OpenCV · Streamlit · WebRTC · Grad-CAM · iTunes API

- Developed an facial emotion recognition system on 35K+ images, achieving 69.2% test accuracy across 7 emotion classes while improving minority-class recall by 20 percentage points.
- Engineered a real-time recommendation engine using probabilistic emotion fusion, confidence-based filtering, and feedback-driven personalization to deliver adaptive music recommendations.

EDUCATION

Atria University  *B.Tech in Computer Science (Digital Transformation)*

11/2021 – 08/2025 | Bangalore, India

- CGPA:** 9.02 / 10.0
- Relevant Coursework:** Machine Learning, Deep Learning, NLP, Computer Vision, Big Data, MLOps, Cloud Computing, DSA